

Optimal Sensor Placement in Sewerage Networks: A Literature Review

Rijoy John (a1961794)

Adelaide University, South Australia

Abstract. Wastewater and stormwater collection systems are among the largest, oldest, and least observable pieces of urban infrastructure, and every additional monitoring sensor is expensive to procure, install, and maintain. This has motivated a substantial body of research on the *optimal sensor placement* (OSP) problem: given a fixed sensor budget, where should sensors be deployed to maximize the value of the resulting monitoring data? This review synthesizes 16 sources – peer-reviewed journal articles, conference and workshop papers, and one unpublished manuscript – addressing sensor placement, sensor-network design, and related monitoring-data analytics in sewerage and urban drainage networks. The literature is organized into three thematic clusters: meta-heuristic and combinatorial optimization approaches (genetic algorithms, greedy and information-theoretic formulations, Bayesian and sparse-sensing methods); machine learning and graph neural network (GNN) based approaches that jointly address placement and state reconstruction; and GIS-based and risk/multi-criteria decision frameworks. Cross-cutting trends identified include convergence toward graph/topology-aware methods, growing attention to city-scale computational feasibility, diversification of monitoring objectives (pandemic surveillance, blockage/overflow detection, inflow and infiltration detection, flood-risk monitoring, asset-failure risk management), and an emerging trend of jointly optimizing placement with hydraulic model calibration or reconstruction accuracy. A dedicated synthesis further shows that most surveyed methods assume a sensor’s detection capability is uniform across candidate locations, with only two studies explicitly modeling detection capability as position-dependent on local network topology. The review closes by identifying six open research gaps – most notably the lack of standardized evaluation metrics and the scarcity of position-dependent detection modeling – to inform the direction of a capstone project on sensor placement in sewerage networks.

1 Introduction

Wastewater and stormwater collection systems are among the largest, oldest, and least observable pieces of urban infrastructure. Continuous in-sewer monitoring – using flow meters, water-level sensors, and water-quality probes – is the most direct way to detect anomalies such as blockages, overflows, inflow and infiltration (I&I), and disease outbreaks reflected in wastewater. However, every additional sensor is expensive to procure, install, power, and maintain,

so utilities can typically instrument only a small fraction of an otherwise vast pipe network (Moradi & Najafi 2026, Crowley et al. 2025). This has motivated a substantial body of research on the *optimal sensor placement* (OSP) problem: given a fixed sensor budget (or a target level of observability), where in a sewer network should sensors be deployed to maximize the value of the resulting monitoring data?

This review synthesizes 16 sources – peer-reviewed journal articles, conference and workshop papers, and one unpublished manuscript – that address sensor placement, sensor-network design, and related monitoring-data analytics in sewerage and urban drainage networks. The literature is organized into three thematic clusters that reflect the dominant methodological paradigms in the field:

1. **Metaheuristic and combinatorial optimization approaches** (Section 2) – greedy, genetic, information-theoretic, and Bayesian formulations of the placement problem.
2. **Machine learning and graph neural network (GNN) based approaches** (Section 3) – data-driven methods that jointly address placement and state reconstruction/prediction at unmonitored locations.
3. **GIS-based and risk/multi-criteria decision frameworks** (Section 4) – spatial and risk-weighted approaches that incorporate asset condition, geography, and failure likelihood into placement decisions.

One review article that summarizes the broader field is discussed separately in Section 5, and Section 6 synthesizes cross-cutting trends, examines how each study models a sensor’s detection capability, compares the surveyed methods, and identifies open research gaps, before Section 8 concludes.

2 Metaheuristic and Combinatorial Optimization Approaches

The largest cluster of surveyed papers formulates OSP as a combinatorial or continuous optimization problem, typically solved with greedy heuristics, genetic algorithms (GAs), or information-theoretic objective functions.

2.1 Genetic and evolutionary algorithms

Several studies adapt genetic-algorithm and NSGA-II (Non-dominated Sorting Genetic Algorithm II) formulations, originally developed for drinking-water distribution network sensor placement, to the sewer domain. Salem & Abokifa (2024) propose a hybrid framework that couples a multilayer-perceptron-based machine-learning source-identification (SI) model with a genetic algorithm, optimizing sensor placement in a real mid-size sewer network to jointly maximize *observability* and *reliability* of wastewater-based source identification – revealing a clear trade-off between these two objectives. Wang, Xu & Zhou (2024) address the related problem of city-scale pandemic (e.g., COVID-19) surveillance

via sewage sampling, introducing a connectivity-based objective evaluation approach embedded in an NSGA-II algorithm; applied to a real network serving over 500,000 residents in Hong Kong, the method outperformed human placement heuristics by 102%. Their sensing-coverage objective is a priority-weighted sum corrected for upstream–downstream overlap, given by Eq. (1):

$$\text{Coverage} = \sum_{k=1}^K \delta_k \sum_{j \in \text{cov}(k)} w_j, \quad (1)$$

where w_j is a location-specific priority score (population size or pandemic-risk factor) assigned to manhole j , and the connectivity indicator $\delta_k \in \{0, 1\}$ marks whether sensor k 's catchment is not already subsumed by another sensor further downstream – so that overlapping upstream areas are not double-counted and the total coverage of a sensor set is not simply the sum of each sensor's individual coverage.

Ninh et al. (2025) instead combine a GA with a topology- and elevation-derived *blockage detection condition* to determine the minimum number and optimal layout of sensors needed to cover two real gravity sanitary sewer systems across multiple investment phases, directly targeting overflow and blockage detection rather than source identification. Formally, a candidate location s satisfies their detection condition only if Eq. (2) holds:

$$Z_0 + H_d > Z_s + H_s, \quad (2)$$

where Z_0 and H_d are the invert elevation and overflow-triggering depth threshold (set to 70% of full pipe depth) at the sensor's downstream node, and Z_s and H_s are the invert elevation and typical dry-weather depth at the candidate location itself – an explicitly position-dependent inequality that is precisely why coverage capacity varies location by location within the same sewer system.

Wang, Xia, Chen, Tong & Zhou (2024) revisit the pandemic-surveillance placement problem addressed by Wang, Xu & Zhou (2024), proposing an *Evolutionary Greedy* (EG) algorithm designed for computational efficiency on large-scale, directed sewage networks. Benchmarked on both synthetic small-scale networks and the same Hong Kong sewage network, EG achieves consistent efficiency gains while maintaining solution quality comparable to more expensive multi-objective searches, illustrating the field's ongoing push toward scalability as candidate networks grow to tens of thousands of nodes.

2.2 Information-theoretic and greedy heuristic formulations

Crowley et al. (2025) depart from GA-based search, instead posing sensor placement as an information-maximization problem: using a stochastic description of network states, *mutual information* is proposed as the measure of evidence gained about network state from a candidate sensor configuration. To circumvent the resulting combinatorial explosion, the authors introduce a *one-step modified greedy*

algorithm that explores solutions beyond the standard greedy trajectory within tractable computation time, validated on two real sewer networks (479 and 1,020 nodes respectively) including the well-known Bellinge (Denmark) benchmark network. Their placements are shown to outperform other heuristics when validated against linear and neural-network estimators of unmonitored locations – an evaluation protocol that foreshadows the reconstruction-based validation used throughout the GNN literature in Section 3.

Moradi & Najafi (2026) present a comprehensive *Descriptive–Predictive–Prescriptive Sensor-Placement* (DPP-SP) framework that links machine-learning failure prediction (Random Forest, XGBoost, and a multilayer perceptron trained on 2020–2025 pipe-failure data, with Random Forest achieving the best ROC-AUC of 0.7626) to a *risk-weighted maximum-coverage* placement problem, solved via greedy, GA, and tabu-search optimizers over 680 candidate sites in a 33,349-pipe network. Each pipe’s risk score is defined multiplicatively, Eq. (3):

$$r_i = p_i \cdot s_i, \quad (3)$$

combining the ML-predicted failure probability p_i with a diameter-derived consequence-severity weight s_i (Section 7.2 discusses this weight further), and every placement is scored against a single fixed-reference coverage ratio, Eq. (4):

$$\text{Coverage}(\%) = \frac{\sum_{i \in \text{covered}} r_i}{\text{REF}_{\text{TOTAL}}} \times 100, \quad (4)$$

so that models with different, uncalibrated probability scales remain directly comparable. All three optimizers converge on the same solution at a budget of 48 sensors, achieving 32.26% risk coverage versus an 11.73% status-quo baseline (a 174.9% improvement), while a sensitivity study identifies detection range as the most influential design parameter. This paper is notable both for its optimization methodology and its explicit risk weighting, bridging the metaheuristic and risk-based literatures (see also Section 4).

2.3 Bayesian and sparse-sensing formulations

Huang et al. (2025) tackle the coupled problem of sensor placement *and* hydraulic model calibration, arguing that in practice sensors are often unavailable precisely when model calibration is needed. Their method first applies a graph-partition strategy to ensure spatial uniformity of candidate sensors, then uses *Bayesian Experimental Design* – maximizing expected data worth via the relative entropy between prior and posterior parameter distributions – to select sensor locations, followed by sequential model calibration with an ensemble smoother. Tested on two real urban drainage models under varying rainfall and parameter scenarios, the joint approach outperforms empirical placement strategies (even sensor distribution, or placement at known flood hotspots) in both prediction accuracy and uncertainty reduction.

Ding et al. (2025) propose a *data-driven sparse sensing* (DSS) approach grounded in linear-algebraic model reduction rather than combinatorial search: EPA-SWMM

simulation ensembles are used to build a reduced basis via singular value decomposition (SVD), and QR-factorization-based sensor selection identifies the most informative subset of candidate locations for system-level flow reconstruction. Demonstrated on a digital twin of the Woodland catchment (Duluth, Minnesota, USA), just three sensors among 77 candidates achieve a mean Nash–Sutcliffe efficiency of 0.949 across observed storm events, closely tracking the exhaustive-search optimum while substantially outperforming Monte Carlo random placement – underscoring that principled subset-selection methods can approach optimality at a fraction of the search cost of combinatorial heuristics.

3 Machine Learning and Graph Neural Network (GNN) Based Approaches

A second, rapidly growing cluster reframes sensor placement around *data-driven state reconstruction*: rather than (or in addition to) searching for an optimal location set directly, these papers train graph-aware machine learning models that infer the hydraulic state at unmonitored nodes from a sparse set of monitored ones, and use the resulting predictive utility to evaluate or guide placement.

He, Nan, Chen, Ye, Ji, Chen, Zhang, Wu, Liu, Ge & Che (2025) propose a spatiotemporal graph convolutional network (STGCN) that reconstructs water-level data at unmonitored urban drainage nodes using only sparse monitoring data and network topology; on a case-study network, water levels at 527 unmonitored nodes were reconstructed from only seven monitoring nodes, achieving 71.3% accuracy with a median mean absolute error of 0.038 m, after tuning spatial data dimensionality, a passive-masked training strategy, and historical input length. Building on this, He, Nan, Ye, Chen, Ji, Chen & Xiao (2025) introduce an *edge-attribute-enhanced* STGCN (Edge-STGCN) that incorporates physical pipe attributes (notably pipe invert elevation) rather than relying on simple topological connectivity alone; with only 10% of nodes monitored, Edge-STGCN achieves reliable predictions at 88.6% of system-wide nodes, substantially outperforming a plain multilayer perceptron (14.5%) and a topology-only STGCN (47.0%), and identifies pipe invert elevation as a key driver of prediction accuracy – particularly for small-diameter branches near the outfall. He et al. (2026) extend this line of work into an explicit end-to-end *sensor-placement* framework: a *topology-first and demand-driven* approach introduces the notion of an *ideal perceptual domain* (IPD) to quantify the theoretical perceptibility achievable by a candidate sensor configuration a priori, guiding placement toward a target coverage level, while a graph-based reconstruction model estimates the resulting *actual perceptual domain* (APD). In a case study, targeting 90% IPD coverage required sensors at only 3.0% of network nodes, yet achieved 86.7% APD coverage during a representative rainfall event with an IPD–APD consistency coefficient of 0.96 and flood-node identification precision/recall of 0.95/0.91 – directly linking a theoretical placement-quality metric to real-time flood-risk monitoring performance.

Chen & Yu (2025) target Low Impact Development (LID) stormwater systems specifically, proposing a *cluster-then-forecast* framework: a customized clustering algorithm identifies representative centroid nodes for sensor placement, after which an LSTM model uses the preceding 25 minutes of water-level data at monitored nodes to forecast system-wide water levels 5 minutes ahead, including at unmonitored nodes – addressing the practical problem of shifting flood hotspots under nonstationary rainfall and urbanization. Qi et al. (2025) take a purely *unsupervised* approach to monitoring-point placement, combining a graph convolutional network for topological feature extraction with a structured deep clustering network to identify critical monitoring zones without labeled training data; validated on a self-collected dataset and the public C-Town benchmark, the method achieves over 94% clustering accuracy against expert-defined labels, reduces the number of required monitoring points by over 40% while increasing critical-node coverage above 90%, and shortens simulated pollution-event response times by nearly 36%.

Finally, Wang et al. (2026) address a distinct monitoring objective – inflow and infiltration (I&I) detection – with a *topology-derived flow inference* (TDFI) method that synthesizes information from multiple upstream sensors (rather than relying on isolated, fixed detection thresholds per sensor) via a novel Detection Threshold metric, optimized using a deterministic Sequential Backward Selection algorithm. The framework shows that prioritizing sensor density in high building-density areas significantly improves I&I detection performance and scales robustly to both synthetic and large real-world networks, demonstrating that graph/topology-aware reasoning is valuable even without a full deep-learning architecture.

4 GIS-Based and Risk/Multi-Criteria Decision Frameworks

A second strand of the literature situates sensor placement within broader spatial and risk-assessment frameworks, often predating or complementing the algorithmic optimization literature above. Nguyen et al. (2021) apply a GIS-based fuzzy Analytical Hierarchy Process (AHP) to delineate suitable areas for flood-monitoring sensor placement in the wastewater/stormwater network of Ålesund, Norway. Eight criteria – including altitude, rainfall, geology, manhole locations, population density, critical infrastructure proximity, road network, and traffic load – are weighted via fuzzy AHP and combined in a GIS environment to produce suitability maps; the analysis identifies manholes, altitude, and rainfall as the most sensitive factors for sensor placement. Unlike the optimization-driven papers in Section 2, this approach does not optimize an explicit objective function but instead produces a multi-criteria suitability surface intended to guide expert placement decisions.

This suitability-scoring paradigm contrasts instructively with the risk-driven DPP-SP framework of Moradi & Najafi (2026) (Section 2), which likewise in-

corporate non-hydraulic criteria into placement but does so by weighting an explicit maximum-coverage *objective function* with machine-learning-predicted pipe-failure probabilities, rather than by producing an expert-facing suitability surface. Together, the two studies illustrate two distinct ways multi-criteria information (geography, asset risk) can enter a placement decision: as a qualitative screening layer prior to optimization, or as a quantitative weight inside it.

5 Review and Overview Articles

One of the surveyed sources is itself a literature review, offering a useful cross-check on the thematic structure adopted here. Moradi et al. (2025) provide a broad overview of sensor-network optimization and data analytics for smart wastewater and stormwater collection systems, surveying heuristic, metaheuristic, and reinforcement-learning optimization techniques alongside AI-driven predictive analytics, and illustrating (via case examples) how these tools reduce maintenance costs and improve regulatory compliance and infrastructure resilience. It echoes an observation borne out across this literature review: quantitative evaluation of monitoring-scheme quality remains inconsistent across studies, and existing layout methods are rarely demonstrably optimal, motivating continued research into standardized evaluation metrics and dynamic, phased-deployment strategies.

6 Synthesis and Research Gaps

6.1 Cross-cutting trends

Several trends emerge across the 16 surveyed sources:

- **Convergence toward graph/topology-aware methods.** Whether via GA-based search over network topology (Ninh et al. 2025), GNN-based reconstruction (He, Nan, Chen, Ye, Ji, Chen, Zhang, Wu, Liu, Ge & Che 2025, He, Nan, Ye, Chen, Ji, Chen & Xiao 2025, He et al. 2026, Qi et al. 2025), or topology-derived inference (Wang et al. 2026), the network’s graph structure is increasingly treated as a first-class input rather than an implicit constraint.
- **Growing attention to scalability.** Several studies explicitly target large, real-world, city-scale networks (tens of thousands of pipes/nodes) and benchmark computational efficiency alongside solution quality (Wang, Xu & Zhou 2024, Wang, Xia, Chen, Tong & Zhou 2024, Moradi & Najafi 2026, Crowley et al. 2025), reflecting the field’s maturation from proof-of-concept case studies toward deployment-scale feasibility.
- **Diversification of monitoring objectives.** While early OSP research targeted generic hydraulic observability, recent studies target specific outcomes: pandemic/wastewater surveillance (Salem & Abokifa 2024, Wang, Xu & Zhou 2024, Wang, Xia, Chen, Tong & Zhou 2024), blockage and overflow

detection (Ninh et al. 2025), inflow/infiltration detection (Wang et al. 2026), flood-risk monitoring (Chen & Yu 2025, He et al. 2026), and asset-failure risk management (Moradi & Najafi 2026).

– **Joint treatment of placement and model calibration/reconstruction.**

A growing subset of papers no longer treats placement as a standalone problem but jointly optimizes it with hydraulic model calibration (Huang et al. 2025) or downstream state-reconstruction accuracy (He, Nan, Chen, Ye, Ji, Chen, Zhang, Wu, Liu, Ge & Che 2025, He, Nan, Ye, Chen, Ji, Chen & Xiao 2025, He et al. 2026, Ding et al. 2025), suggesting the field is moving from “where to place sensors” toward “how to design an end-to-end sensing-and-inference system.”

6.2 Sensor detection capability: homogeneous vs. position-dependent modeling

A dimension that cuts across the clusters above – but is rarely discussed explicitly by the papers themselves – is *how each study models a sensor’s own ability to detect the phenomenon of interest at a given candidate location*. Four distinct patterns emerge from the surveyed methodologies.

Homogeneous detection/measurement models (the majority default). Most surveyed studies implicitly or explicitly assume that a sensor’s detection fidelity is uniform across every candidate location, so that placement quality is driven entirely by *where* sensors are placed relative to the network, not by any location-specific variation in *how well* a sensor can sense there. Two papers make this assumption explicit and quantitative: Huang et al. (2025) assign every candidate water-level sensor the same Gaussian observation-error model ($\mathcal{N}(0, 0.1 \text{ m})$) regardless of location, and Ding et al. (2025) likewise apply identical Gaussian measurement noise to every sensor in their SVD/QR sparse-sensing formulation. Moradi & Najafi (2026) apply a single, network-wide *detection radius* $L_{\max}$ as the design parameter governing which pipes a sensor can observe – but the authors explicitly flag this as a simplification, noting that real detection range depends on sensor technology (level/flow instruments reaching 1000–1500 m vs. acoustic/water-quality methods confined to 200–500 m) and on local hydraulics (self-cleansing velocity, junction density, wet- vs. dry-weather flow), concluding that “the single fixed radius adopted here is best interpreted as a spatially averaged design value rather than a physical constant.” The remaining optimization- and ML-oriented papers (Salem & Abokifa 2024, Crowley et al. 2025, Wang, Xu & Zhou 2024, Wang, Xia, Chen, Tong & Zhou 2024, Chen & Yu 2025, He, Nan, Chen, Ye, Ji, Chen, Zhang, Wu, Liu, Ge & Che 2025, He, Nan, Ye, Chen, Ji, Chen & Xiao 2025, He et al. 2026, Qi et al. 2025) do not discuss location-dependent detection fidelity at all; a sensor is implicitly treated as returning an exact or uniformly-noisy reading of its own node.

Position-dependent detection capability, derived from topology. Exactly two of the surveyed papers explicitly model a candidate location’s detection capabil-

ity as a function of the surrounding network structure, rather than as a fixed, location-independent property. Ninh et al. (2025) is the clearest example: their *blockage detection condition* computes a location-specific *coverage capacity* directly from network topology and elevation data, so that two candidate manholes in the same system can have materially different detection ability depending on their upstream pipe layout – the paper reports this coverage capacity varying substantially between manholes within the same sewer system. Wang et al. (2026) make an equivalent point for inflow and infiltration detection: they explicitly critique prior work’s reliance on “fixed, isolated detection thresholds derived from individual sensors,” and instead compute a pipe-specific *Individual Pipe Detection Threshold* (IPDT) by synthesizing information from multiple upstream sensors via network topology. Their case-study network is motivated precisely by this heterogeneity: one sub-area covering 16.7% of the service area contributes roughly 32% of daily flow, a hydraulic imbalance the authors argue is exactly the kind of “heterogeneous flow pattern” that a fixed, location-independent detection threshold would fail to capture. These two papers are the only sources in the review that treat detection capability itself – not merely placement value – as position-dependent.

Homogeneous sensing, but position-dependent informational value. A related but conceptually distinct form of heterogeneity appears throughout the ML/GNN cluster (Section 3): the sensor itself is still assumed to report its own node accurately, but the *value* of placing it at a given node – measured as the resulting gain in system-wide reconstruction accuracy, forecast quality, or population coverage – varies with network position. Ding et al. (2025) illustrate this most precisely: alongside their homogeneous noise model, they separately quantify a *heterogeneous dropout sensitivity* across sensor positions, attributing most of the variation in reconstruction-accuracy loss upon sensor removal to structural factors (pipe length contributes the largest share, 64%, followed by QR selection order at 22%). Wang, Xu & Zhou (2024) and Wang, Xia, Chen, Tong & Zhou (2024) similarly assign each candidate node a position-dependent *value* via a connectivity-based objective tied to the size of the upstream population or catchment served, rather than via any change in detection fidelity. This is a meaningfully different axis of heterogeneity from the topology-derived *detection* models above: it varies how much a placement is *worth*, not how well a sensor *senses*.

Not a detection model at all. Nguyen et al. (2021) sits outside this taxonomy: the fuzzy-AHP suitability score is not a model of sensor detection performance in any sense, but a pre-placement desirability score derived from geographic and demographic criteria (altitude, rainfall, population density, etc.).

Only 2 of the 15 optimization/ML papers surveyed in Section 6.3 formalize position-dependent detection capability, against 3 that use an explicit but homogeneous measurement model and roughly 10 that leave the assumption implicit. This imbalance is carried forward as Research Gap 6 below.

6.3 Comparison of methods

Table 1 summarizes the surveyed optimization and learning-based studies by primary method, monitoring objective, target network type, and case-study scale.

Table 1: Comparison of sensor-placement methods across the surveyed literature.

Source	Primary method	Monitoring objective	Network type	Case-study scale
Moradi & Najafi (2026)	ML risk prediction + greedy/GA/tabu search	Risk-weighted failure coverage	Sanitary sewer	33,349-pipe system
He et al. (2026)	Topology-first IPD/APD framework + GNN	Hydraulic-state reconstruction, flood risk	Urban drainage (general)	Sensors at 3.0% of nodes
Wang et al. (2026)	Topology-derived flow inference + Sequential Backward Selection	Inflow & infiltration detection	Sanitary sewer	Synthetic + large real network
Crowley et al. (2025)	Mutual information + modified greedy	Generic hydraulic observability	Sewer (general)	479–1020 node real networks
Ninh et al. (2025)	GA + topology/elevation blockage rule	Blockage / overflow detection	Sanitary sewer (gravity)	2 real gravity sewer systems
Huang et al. (2025)	Bayesian experimental design + graph partition	Joint placement & model calibration	Urban drainage (general)	2 real UDMs
Ding et al. (2025)	SVD + QR-factorization sparse sensing	Flow reconstruction	Stormwater / urban drainage	Digital twin, 77 candidate nodes
Chen & Yu (2025)	Clustering + LSTM forecasting	Real-time LID water-level prediction	Stormwater (LID)	LID stormwater case study
He, Nan, Chen, Ye, Ji, Chen, Zhang, Wu, Liu, Ge & Che (2025)	Spatiotemporal GCN (STGCN)	Water-level reconstruction	Urban drainage (general)	527 unmonitored nodes, 7 sensors
He, Nan, Ye, Chen, Ji, Chen & Xiao (2025)	Edge-attribute-enhanced STGCN	Nodal water-level prediction	Urban drainage (general)	Sparsely monitored UDN (10% nodes)
Qi et al. (2025)	GCN + structured deep clustering	Monitoring-point placement	Urban drainage (general)	Self-collected data + C-Town benchmark
Salem & Abokifa (2024)	MLP + genetic algorithm	Wastewater source identification	Sanitary sewer	Mid-size real sewer network

Source	Primary method	Monitoring objective	Network type	Case-study scale
Wang, Xu & Zhou (2024)	NSGA-II (multi-objective GA)	Pandemic sewage surveillance	Sanitary sewer	500,000-resident network (Hong Kong)
Wang, Xia, Chen, Tong & Zhou (2024)	Evolutionary greedy (EG) algorithm	Pandemic sewage surveillance	Sanitary sewer	Same Hong Kong network
Nguyen et al. (2021)	GIS + fuzzy AHP (multi-criteria)	Flood-sensor suitability mapping	Combined (wastewater + stormwater)	Ålesund, Norway

6.4 Research gaps

Based on the surveyed literature, several gaps are apparent:

1. **Lack of standardized evaluation metrics.** As noted by Moradi et al. (2025), studies use heterogeneous objective functions (mutual information, coverage percentage, reconstruction accuracy, risk-weighted coverage), which makes cross-study comparison difficult and slows convergence on best practice.
2. **Limited validation on very large, real, multi-purpose networks.** While several studies use real networks of meaningful scale (Crowley et al. 2025, Wang, Xu & Zhou 2024, Moradi & Najafi 2026), most GNN-based reconstruction studies (Section 3) are validated on a single case-study catchment, leaving open questions of transferability across network topologies, climates, and pipe-material distributions.
3. **Sparse treatment of multi-objective, multi-purpose placement.** Most studies optimize for a single downstream purpose (e.g., I&I detection *or* flood risk *or* pandemic surveillance); few address the realistic utility scenario in which a single sensor network must simultaneously serve several of these purposes under one shared budget.
4. **Dynamic and phased deployment remains underexplored.** Among the surveyed papers, only Moradi & Najafi (2026) and Ninh et al. (2025) explicitly consider multi-phase or incremental investment; dynamic optimization through phased deployment and feedback iteration otherwise remains a largely open direction, with adaptive placement needing further attention as networks, budgets, and risk profiles evolve over time.
5. **Scarcity of open benchmark datasets.** Several studies (Wang, Xu & Zhou 2024, He, Nan, Chen, Ye, Ji, Chen, Zhang, Wu, Liu, Ge & Che 2025) explicitly release code and data to support reproducibility, but this remains

the exception; wider adoption of shared, public benchmark networks (in the spirit of the Bellinge and C-Town datasets used by Crowley et al. (2025) and Qi et al. (2025) respectively) would strengthen comparability across the field.

6. **Position-dependent detection modeling is rare.** As shown in Section 6.2, only Ninh et al. (2025) and Wang et al. (2026) explicitly model a sensor’s detection capability as a function of local network topology; the rest of the surveyed literature either assumes a homogeneous, location-independent measurement model or leaves the assumption unstated. Given that Moradi & Najafi (2026) explicitly identify detection range as the most influential design parameter in their own sensitivity study, treating detection capability as spatially uniform may materially bias placement recommendations in networks with heterogeneous hydraulics – a largely unaddressed methodological gap.

7 Possible Contributions and Research Directions

7.1 Motivation: sensor placement under network growth

None of the 16 surveyed sources targets the specific scenario motivating the capstone project behind this review: choosing sensor locations so that observability keeps pace as new housing developments are connected to an already-built sewer network, progressively consuming the hydraulic headroom of downstream pipes. Ninh et al. (2025) comes closest in spirit – their topology- and elevation-derived *coverage capacity* is, to date, the only sensor-behavior model in this review that varies with a location’s position in the network rather than being fixed (Section 6.2) – but it targets blockage detection under the network’s *current* configuration, not the capacity stress introduced by *future* demand growth.

7.2 Capacity- and diameter-aware modeling already present in the literature

Several surveyed papers already fold pipe capacity, or its physical determinants, into their methodology, though none use it for the growth-driven objective above. Moradi & Najafi (2026) draw an eight-feature per-pipe vector directly from a utility GIS asset register – diameter, age, slope, length, material, pipe class, and upstream/downstream node degree – and, notably, use diameter as the primary determinant of the consequence-severity weight s_i in their risk score (Section 2, Eq. (3)) – $s_i \in \{1, 2, 3\}$ for pipes below 203 mm, 203–457 mm, and 457 mm and above or any force main, respectively – even though diameter itself is only a weak predictor of failure *probability* p_i in their model. Wang et al. (2026) go furthest in operationalizing hydraulic capacity directly: for unmonitored pipes, their Individual Pipe Detection Threshold is capped at the pipe’s *full-pipe capacity*, computed via the Manning equation, Eq. (5):

$$Q = \frac{D^{8/3} I^{1/2}}{4^{5/3} n}, \quad (5)$$

from diameter, slope, and roughness – and their case-study motivation explicitly cites “undersized diameters and insufficient conveyance capacity” under “rapid urbanization” as a driver of the flooding the paper aims to address. He, Nan, Ye, Chen, Ji, Chen & Xiao (2025) and He et al. (2026) both use pipe diameter (alongside length and invert elevation) as an edge attribute in their GNN reconstruction models – but He et al. (2026) explicitly note that these physical attributes feed only the reconstruction model and are “not incorporated into the subsequent sensor-placement” decision itself, leaving a paper-acknowledged gap between capacity-relevant data already being collected and its use in placement.

7.3 A proposed direction for the capstone contribution

Taken together, the surveyed literature supplies two ingredients this direction would need to combine, but no source combines them: a position-dependent detection/coverage-capacity model tied to network topology (Ninh et al. 2025, Wang et al. 2026), and a diameter-derived hydraulic capacity signal extracted from GIS pipe attributes already shown to be predictive elsewhere (Moradi & Najafi 2026, Wang et al. 2026). A concrete formulation would (i) compute each pipe’s full-pipe hydraulic capacity $Q_{\text{cap},i}$ from diameter, slope, and roughness in the manner of Wang et al. (2026); (ii) inject a hypothesized additional demand at the node(s) representing a planned housing development and propagate the resulting flow increase $\Delta Q_{d,i}$ downstream through the network graph; (iii) rank pipes by the resulting drop in capacity headroom to flag likely future chokepoints; and (iv) place sensors to maximize position-dependent observability, in the spirit of Ninh et al. (2025), at those chokepoints under a fixed sensor budget, using a headroom-weighted coverage search analogous to Moradi & Najafi (2026)’s risk-weighted formulation but with growth-driven headroom loss substituted for failure probability as the weighting signal.

Formally, this substitutes a headroom-consumption ratio, Eq. (6):

$$h_i = \frac{\Delta Q_{d,i}}{Q_{\text{cap},i}}, \quad (6)$$

for the failure-probability term in Moradi & Najafi’s risk score (Eq. (3)), while keeping a diameter- or criticality-derived consequence weight analogous to their s_i ; a candidate sensor location s would count toward covering pipe i only if it satisfies a growth-adapted version of Ninh et al.’s detection condition (Eq. (2)), with the downstream depth threshold H_d reinterpreted as the depth at which headroom is exhausted rather than at which overflow occurs; and the resulting per-location values could be aggregated with the priority-weighted, overlap-corrected coverage sum of Wang, Xu & Zhou (Eq. (1)), substituting h_i for the population-based priority score w_j so that overlapping downstream catchments of a shared development are not double-counted. This would simultaneously address the review’s Research Gaps 3 (multi-objective placement), 4 (dynamic/phased deployment, since growth unfolds incrementally), and 6 (position-dependent detection modeling) – none of which is fully resolved by the current literature – while

targeting a placement driver, anticipated capacity exhaustion from network expansion, that is entirely absent from the 16 sources reviewed here.

8 Conclusion

The 16 sources reviewed here span increasingly sophisticated, data-driven optimization frameworks that integrate combinatorial search, information theory, Bayesian experimental design, graph neural networks, and GIS-based multi-criteria scoring. A clear trajectory is visible: contemporary research increasingly treats sensor placement not as an isolated combinatorial problem but as one component of an end-to-end sensing-and-inference pipeline that jointly considers network topology, failure/flood risk, and downstream state-reconstruction or prediction accuracy. At the same time, Section 6.2’s synthesis shows that this sophistication has been applied almost entirely to *where* sensors go, while *how well* a sensor detects at a given location is left as a uniform, largely unexamined assumption in all but two of the surveyed studies. For a capstone project on optimal sensor placement in sewerage networks, this body of work suggests that competitive contributions will most likely come from (a) explicitly stating and justifying the placement objective (observability, risk coverage, reconstruction accuracy, or a stated combination), (b) validating on a real or realistically-scaled network topology, (c) benchmarking against both simple heuristics (e.g., random or degree-based placement) and at least one established method from the categories surveyed above, and (d) considering whether a position-dependent detection-capability model – in the spirit of Ninh et al. (2025) and Wang et al. (2026) – would materially change the resulting placement relative to the homogeneous-detection assumption used elsewhere in the field.

Bibliography

- Chen, E. & Yu, X. (2025), ‘A novel machine learning based cluster-then-forecast framework for sensor placement optimization and real-time water level prediction in low impact development (lid) stormwater system’, *ACS EST Water* **5**(7), 3729–3738.
- Crowley, G., Tait, S., Panoutsos, G., Speight, V. & Esnaola, I. (2025), ‘Information-theoretic sensor placement for large sewer networks’, *Water Research* **268**, 122718.
- Ding, Z., Kumar, A., Islam, I. M. A., Montezuma, M. A., Zhang, R. & Zhang, K. (2025), Optimizing sensor placement for flow reconstruction in urban drainage networks: A digital twin-based sparse sensing approach. Manuscript submitted to Water Research.
- He, L., Nan, J., Chen, L. & Ye, X. (2026), ‘A topology-first and demand-driven sensor placement framework for hydraulic-state reconstruction in urban drainage networks’, *Water Research* **307**, 126713.
- He, L., Nan, J., Chen, L., Ye, X., Ji, S., Chen, Z., Zhang, Y., Wu, F., Liu, B., Ge, Z. & Che, Y. (2025), ‘Spatiotemporal graph convolutional network using sparse monitoring data for accurate water-level reconstruction in urban drainage systems’, *Journal of Hydrology* **652**, 132681.
- He, L., Nan, J., Ye, X., Chen, L., Ji, S., Chen, Z. & Xiao, Q. (2025), ‘A graph neural network using physical attributes to improve the system-wide nodal water-level prediction in sparsely monitored urban drainage systems’, *Journal of Hydrology* **663**, 134306.
- Huang, Y., Xiong, J., Zhang, J., Zheng, F., Ji, Y. & Gupta, H. (2025), ‘A robust method to simultaneously place sensors and calibrate parameters for urban drainage pipe system models using bayesian decision theory’, *Engineering Applications of Computational Fluid Mechanics* **19**(1), 2473992.
- Moradi, M. & Najafi, M. (2026), ‘Risk-driven sensor placement in sewer networks: A descriptive–predictive–prescriptive framework’, *Water* **18**(15), 1901.
- Moradi, M., Najafi, M., Lin, Z. & Pan, H. (2025), Advancing sensor network optimization and data analytics for smart wastewater and stormwater collection systems: An overview, in ‘Pipelines 2025’, American Society of Civil Engineers (ASCE), p. 463.
- Nguyen, L. V., Bui, D. T. & Seidu, R. (2021), Identification of sensitive factors for placement of flood monitoring sensors in wastewater/stormwater network using GIS-based fuzzy analytical hierarchy process: A case of study in ålesund, norway, in ‘Proceedings of the International Conference on Innovations for Sustainable and Responsible Mining’, Vol. 108 of *Lecture Notes in Civil Engineering*, Springer Nature Switzerland, pp. 79–97. Case study conducted in 2020.
- Ninh, T. H. D., Do, N. C., Zeng, W. & Lambert, M. F. (2025), ‘Optimal sensor placement in smart sewer systems using network topology and elevation’, *Journal of Water Resources Planning and Management*.

- Qi, Y., Li, Q., Cao, G., Li, G., Zhang, L. & Yang, B. (2025), ‘Unsupervised graph-based deep clustering for optimal monitoring point placement in urban drainage networks’, *Engineering Reports* **7**, e70544.
- Salem, A. K. & Abokifa, A. A. (2024), ‘Sensor placement optimization in sewer networks: Machine learning-based source identification approach’, *Journal of Water Resources Planning and Management* .
- Wang, H., Du, K., Zheng, F., Liang, R., Lu, M., Kapelan, Z. & Savic, D. (2026), ‘A topology-derived flow inference approach to optimize sensor placement for effective inflow and infiltration detection in sewer networks’, *Water Research* **302**, 126140.
- Wang, S., Xia, Y., Chen, H., Tong, X. & Zhou, Y. (2024), Evolutionary greedy algorithm for optimal sensor placement problem in urban sewage surveillance, in ‘Proceedings of the 13th SIGKDD International Workshop on Urban Computing (UrbComp’24)’, ACM, Barcelona, Spain. arXiv:2409.16770.
- Wang, S., Xu, K. & Zhou, Y. (2024), ‘Cost-effective sensor placement optimization for large-scale urban sewage surveillance’, *Sustainable Cities and Society* **103**, 105250.